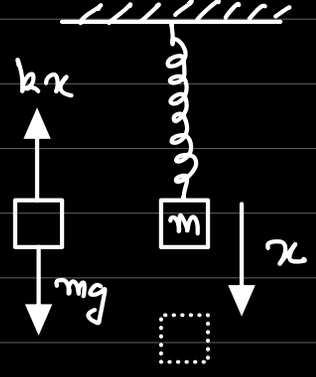
Simple Harmonic Motion:

A motion governed by the differential equation

$$\frac{d^2x}{dt^2} = -\omega^2(x - x_0) \quad (1)$$

is known as Simple harmonic motion.
(Here ω and x_0 are constants.



For instance consider the spring mass system in the figure. For this we can write:

$$mg - kx = m\frac{d^2x}{dt^2}$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}\left(x - \frac{mg}{k}\right)$$ This can be compared to equation (1) with $\omega = \sqrt{\frac{k}{m}}$ and $x_0 = mg/k$.

Here ω is called angular frequency of the motion and x_0 is called mean position.

Solution of the SHM Differential Equation:

Any equation of the form $\frac{d^2x}{dt^2} = -\omega^2(x - x_0)$ is satisfied by following solution forms:

(1) $x - x_0 = A\sin(\omega t + \phi)$ or

(2) $x - x_0 = A\cos(\omega t + \phi)$ where A and ϕ are constants depending on initial velocity and position of the system.

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Verifying the solutions to the SHM DE:

① $x - x_0 = A\sin(\omega t + \phi) \Rightarrow \frac{dx}{dt} = A\omega\cos(\omega t + \phi)$

$\Rightarrow \frac{d^2x}{dt^2} = -A\omega^2\sin(\omega t + \phi) = -\omega^2(x - x_0)$

$\Rightarrow \frac{d^2x}{dt^2} = -\omega^2(x - x_0)$ is automatically satisfied.

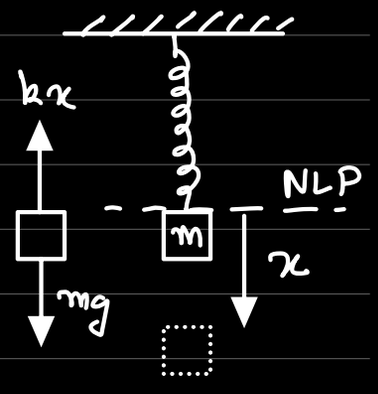
② $x - x_0 = A\cos(\omega t + \phi) \Rightarrow \frac{dx}{dt} = -A\omega\sin(\omega t + \phi)$

$\Rightarrow \frac{d^2x}{dt^2} = -A\omega^2\cos(\omega t + \phi) = -\omega^2(x - x_0)$

$\Rightarrow \frac{d^2x}{dt^2} = -\omega^2(x - x_0)$ is automatically satisfied.

Particular Solution to the SHM DE: When the particular values of A and ϕ are calculated according to given initial conditions, and substituted in the general solution, we get the particular solution. (Typically initial position and initial velocity will be given to solve for A and ϕ).

Example: The spring mass system shown in the figure is released from rest from the natural length position. Calculate the position of the block at any general time.



Sol: $mg - kx = m\frac{d^2x}{dt^2} \Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}\left(x - \frac{mg}{k}\right)$

$\Rightarrow \omega = \sqrt{\frac{k}{m}}, \quad x_0 = mg/k, \quad x(0) = 0, \quad v(0) = 0$

Now let $x - x_0 = A\cos(\omega t + \phi)$ ①

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Now using initial condition, $x(0) = 0$,

$$-x_0 = A\cos\phi \quad (2)$$

Differentiating (1), $v = -A\omega \sin(\omega t + \phi)$, Now use $v(0) = 0$

$$\Rightarrow \quad 0 = -A\omega \sin\phi \quad (3)$$

Now $(2)^2 + ((3)/\omega)^2 \Rightarrow A^2[\sin^2\phi + \cos^2\phi] = x_0^2$

$\Rightarrow A = x_0$ [We always see A as +ve though $-x_0$ is also mathematically ok and equivalent solution would follow].

Now put $A = x_0$ in (2) $\Rightarrow \phi = \pi$

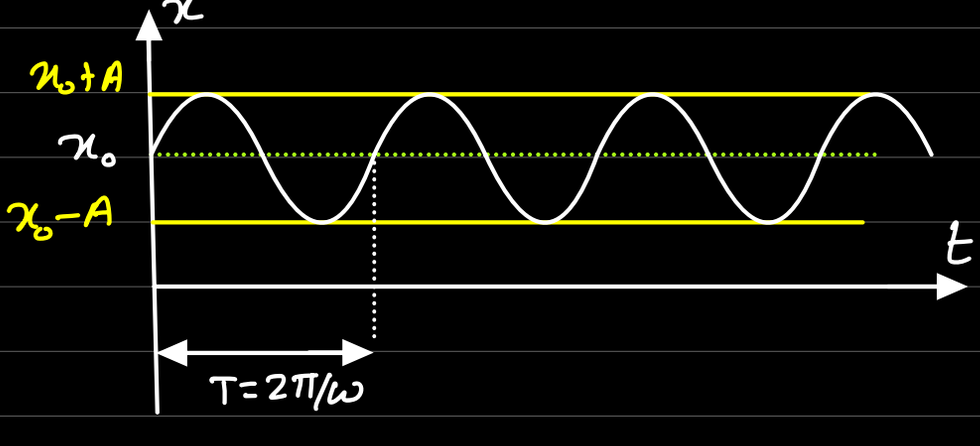
$$\Rightarrow \quad x - x_0 = x_0 \cos(\omega t + \pi)$$

or $x = x_0[1 - \cos(\omega t)]$ where $x_0 = mg/k$, $\sqrt{k/m}$.

Note Very often the origin is taken at the mean position itself $\Rightarrow x_0 = 0$ then our DE becomes: $\frac{d^2x}{dt^2} = -\omega^2(x)$ and the corresponding solutions are:

$$x = A\cos(\omega t + \phi) \text{ or } x = A\sin(\omega t + \phi).$$

Plot of Solution of SHM DE: $x = x_0 + A\sin(\omega t)$



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We see that motion is symmetric about x_0 and hence the word "mean position". Also it can be readily seen that the motion is periodic with a time period given by :

$$\boxed{T = 2\pi/\omega}$$

Also A is the maximum separation of the particle from its mean position and is known as Amplitude.

Frequency of oscillation: The number of cycles executed per unit time is known as frequency of oscillation.

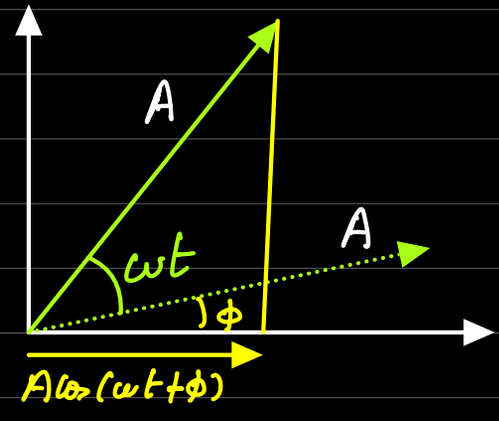
Mathematically it should be the reciprocal of the time period. For instance if each cycle completes in 0.1 sec then no of cycles in 1 sec will be 10 i.e. $\frac{1}{0.1}$.

Thus $\boxed{\nu = \frac{1}{T} = \frac{\omega}{2\pi}}$ Also $\boxed{\omega = 2\pi\nu}$

Phasor Representation of SHM: Any motion of the form

$x = A\cos(\omega t + \phi)$ can be thought of as a shadow of a rotating vector on x axis as shown in the figure.

Here the length of the phasor is A and it starts rotating from an angular position ϕ from x axis rotating with an angular velocity ω anticlockwise.



From the phasor diagram it is clear that the time period of motion must be $T = \frac{2\pi}{\omega}$ because in that much time, the phasor would complete a full circle.

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Example: Two particles are executing SHM about origin along x axis such that their coordinates are given as:

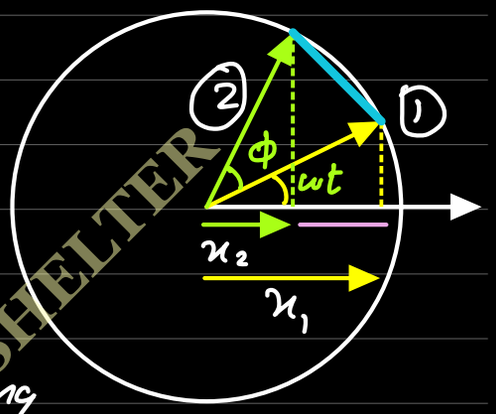
$$x_1 = A\cos\omega t$$

$$x_2 = A\cos(\omega t + \phi)$$

Find (a) maximum separation between the particles

(b) Time at which their separation is maximum for the first time.

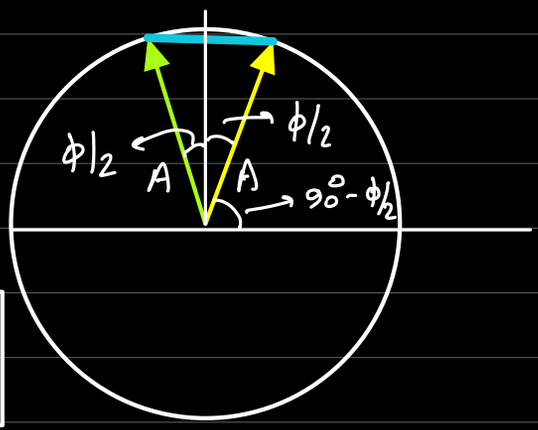
Sol: Consider the phasor diagrams of the two particles at a general time, it can be readily seen that the separation between the particles is just the shadow of the line joining the tip of the two phasors. Now since both phasors are rotating with same angular velocity, the length of this line remains constant and its shadow on x axis is maximum when it is parallel to x axis as shown in the following phasor diagram.



As clear from the adjacent phasor diagram,

max separation, $\Delta x_{max} = 2A\sin(\phi/2)$

and it occurs at a time, $\boxed{t = \dfrac{\pi/2 - \phi/2}{\omega}}$



(∵ the yellow phasor has turned through an angle $90^\circ - \phi/2$.

→ Think Yourself: After how much min time will particles meet each other (and where?! (How would the phasor diagram look like?).

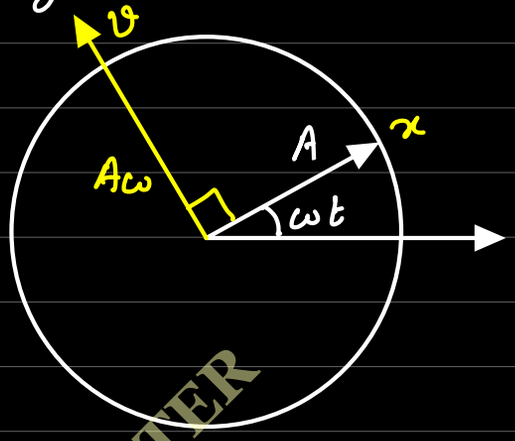
Relation between velocity phasor and position phasor:

Let $x = A\cos(\omega t + \phi) \Rightarrow v = -A\omega\sin(\omega t + \phi)$

$\Rightarrow v = A\omega\cos(\omega t + \phi + \pi/2)$

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Thus we can say that velocity phasor leads the position phasor by a phase of $\pi/2$ and scaled up by a factor of ω. This is graphically shown in the following phasor diagram:



Here "leads" means that velocity attains its maximum positive value at a quarter cycle time before the position phasor attains its maximum positive value. (Both phasors are rotating with ω).

Q. Two particles are oscillating about origin such that their positions are given as:

$$x_1 = 3A\sin(\omega t)$$
$$x_2 = 4A\sin(\omega t + \pi/2)$$

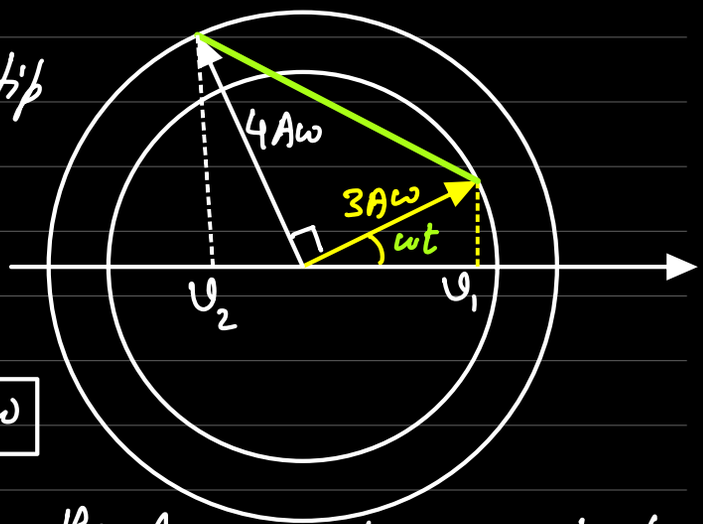
Calculate the maximum relative speed between the two particles and also the time when this happens for the first time.

Sol: $v_1 = 3A\omega\cos(\omega t)$, $v_2 = 4A\omega\cos(\omega t + \pi/2)$, The velocity phasor positions for the two phasors at a general time are shown in the following phasor diagram:

It can be readily seen that relative speed between the particles is given by the projection of the line joining the tip of the two phasors.

Thus maximum relative speed of the particles occurs when this line becomes horizontal and at that time,

$$\boxed{\Delta v_{max} = 5A\omega}$$



(∵ Pythagoras Theorem). Also @ $t=0$, this line is at an angle of 37° with vertical, while at desired instant, it is at 90°

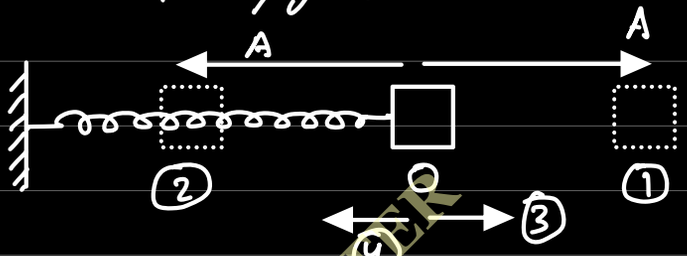
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with vertical $\Rightarrow$ this line has rotated through an angle of 53°. $\Rightarrow$ Time when mean relation used occurs:

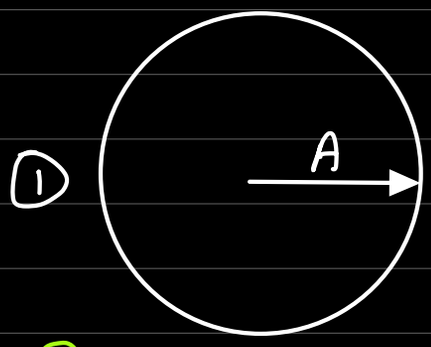
$$t = 53^\circ \times \frac{\pi}{180^\circ} \times \frac{1}{\omega} = \frac{53\pi}{180\omega} \text{ Ans.}$$

Four standard SHM Phasor Positions: Consider a block executing SHM of Amplitude A about O as shown in the figure.

We shall draw the phasors for following standard positions:

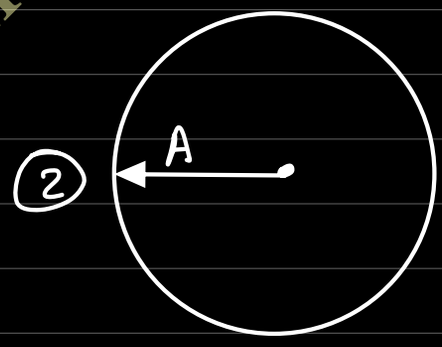


1. Starting from positive extreme @ $t=0$
2. Starting from negative extreme @ $t=0$
3. moving with +ve velocity from mean position @ $t=0$
4. moving with -ve velocity from mean position @ $t=0$



POSITIVE EXTREME

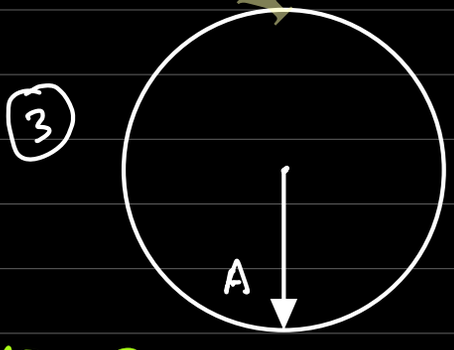
@ $t=0$, $x = +A$, $x = A\cos(\omega t)$



NEGATIVE EXTREME

@ $t=0$, $x = -A$, $x = A\cos(\omega t + \pi)$

$\Rightarrow x = -A\cos(\omega t)$

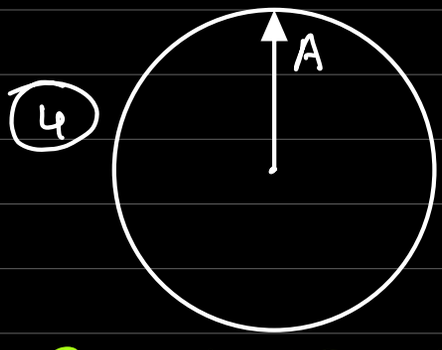


MEAN POSITION WITH +Ve VELOCITY

@ $t=0$, $x=0$, $v>0$

$\Rightarrow x = A\cos(\omega t - \pi/2)$

$\Rightarrow x = A\sin(\omega t)$



MEAN POSITION WITH -Ve VELOCITY

@ $t=0$, $x=0$, $v<0$

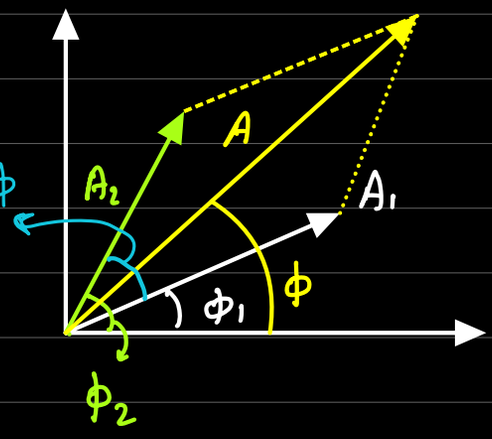
$\Rightarrow x = A\cos(\omega t + \pi/2)$

$\Rightarrow x = -A\sin(\omega t)$

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Addition of Phasors: Let there be two phasors given by:

$x_1 = A_1\cos(\omega t + \phi_1)$ and $x_2 = A_2\cos(\omega t + \phi_2)$ then for purpose of addition we draw the two phasors like vectors at $t=0$ and add them just like vectors to get two quantities



$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\Delta\phi)}$$ and $$\tan\phi = \frac{A_1\sin\phi_1 + A_2\sin\phi_2}{A_1\cos\phi_1 + A_2\cos\phi_2}$$

Now the resultant phasor is given as:

$$x = x_1 + x_2 = A\cos(\omega t + \phi).$$

Similarly if the phasors were in Sines, final phasor will also be reported in sines e.g. if:

$x_1 = A_1\sin(\omega t + \phi_1)$ and $x_2 = A_2\sin(\omega t + \phi_2)$

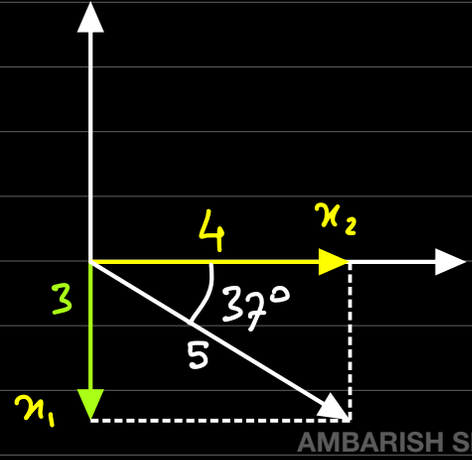
then $x = A\sin(\omega t + \phi)$ where A and ϕ are given by the above formulae.

→ Note: For phasor addition, either both should be converted to sines and final result also reported in sines or both should be converted to cosines and final result also reported in cosines.

Example: Add $x_1 = 3\sin\omega t$ and $x_2 = 4\cos(\omega t)$

Sol: $x_1 = 3\cos(\omega t - \pi/2)$

Refer to the phasor diagram:



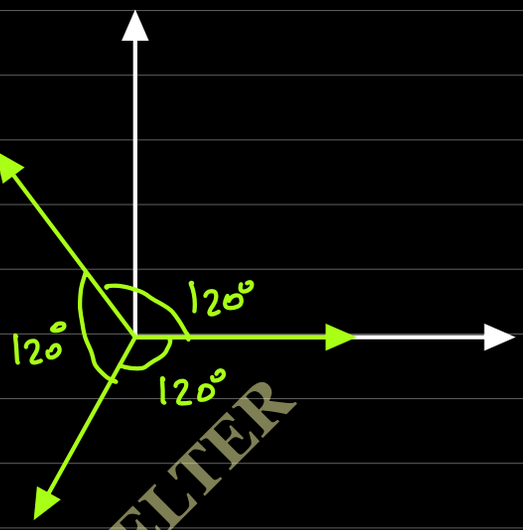
$x = x_1 + x_2 = 5\cos(\omega t - 37°)$

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Example: Add the following SHMs: $x_1 = A\cos\omega t$, $x_2 = A\cos(\omega t + 2\pi/3)$ and $x_3 = A\cos(\omega t - 2\pi/3)$

Sol: Refer to adjacent phasors diagram, Three phasors acting at 120° each would yield a zero resultant by symmetry.

$\therefore x_1 + x_2 + x_3 = 0.$



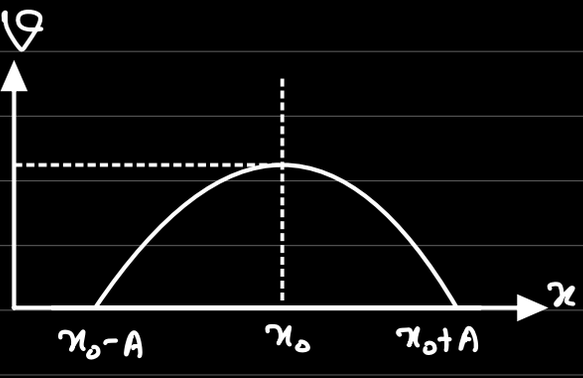
Variation of Velocity with position:

$$x - x_0 = A\cos(\omega t + \phi) \Rightarrow v = -A\omega \sin(\omega t + \phi)$$

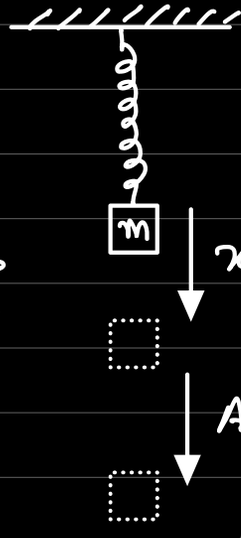
Now $\sin^2\omega t + \cos^2\omega t = 1$

$$\Rightarrow \left(\frac{x - x_0}{A}\right)^2 + \left(\frac{v}{A\omega}\right)^2 = 1 \Rightarrow \boxed{v = \omega\sqrt{A^2 - (x - x_0)^2}}$$

Thus max velocity occurs at $x = x_0$ i.e the mean position and at extreme positions, i.e $x = x_0 + A$ and $x = x_0 - A$, the speed goes to zero.



Energy Variation in SHM: Consider the system shown in the figure. Here total mechanical energy remains constant so it can be evaluated at any position. Let Amplitude of oscillation be A and mean position be located at x_0 below the natural length position then at lower extreme position:



$$U = \frac{1}{2}k(x_0 + A)^2 - mg(x_0 + A) = E$$

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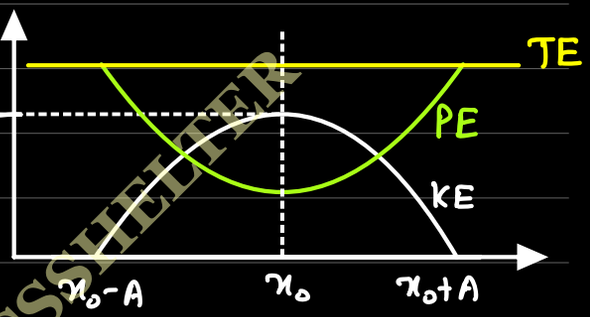
Also KE at a general position is given by, $KE = \frac{1}{2}mv^2$

$\Rightarrow KE = \frac{1}{2}m\left[\omega\sqrt{A^2-(x-x_0)^2}\right]^2 = \frac{1}{2}m\omega^2(A^2-(x-x_0)^2)$

This is an inverted parabola with peak at $x = x_0$.

Also $PE = E - KE$ and $\therefore$ E is constant this must be upward opening parabola with vertex at $x = x_0$.

Here PE depends on reference and $\therefore$ TE also depends on reference where PE is considered as zero. Very often for sake of convenience, PE is taken as zero the mean position.

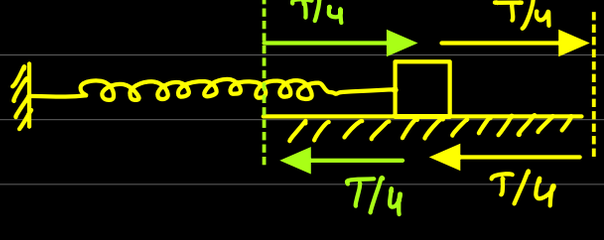


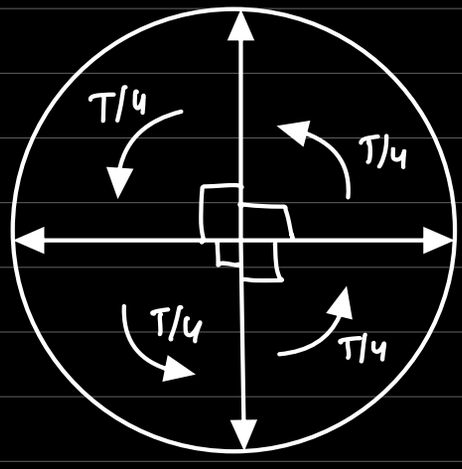
In that case, total energy is same as KE at the mean position, and PE at a general x is then simply,

$PE = \frac{1}{2}m\omega^2A^2 - \frac{1}{2}m\omega^2(A^2-(x-x_0)^2)$

Also $\omega = \sqrt{\frac{k}{m}} \Rightarrow PE = \frac{1}{2}k(x-x_0)^2$ and if mean position is taken as the origin then simply, $PE = \frac{1}{2}kx^2$.

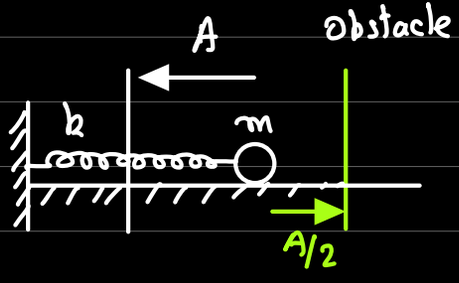
→ Note: SHM is symmetric about the mean position as can be readily seen from phasor diagram.





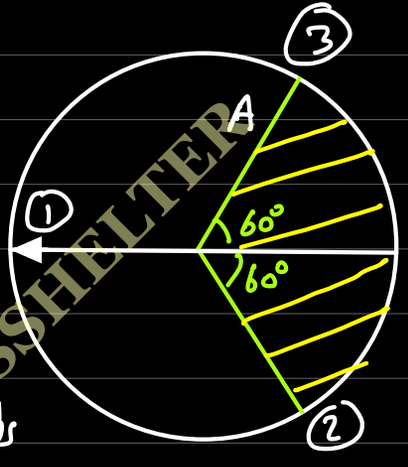
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Clipped SHM (Piecewise SHM):



Q. A spring mass system was performing SHM of amplitude A when an obstacle was placed at a distance $A/2$ from mean position. If collisions with obstacle are perfectly elastic, calculate the time period of resulting oscillations.

Sol. Refer to the adjacent phasor diagram, It can be readily seen that the shaded region of phasor position never occurs due to obstacle at $A/2$ (i.e $A\omega b3$) Thus if the phasor starts from position ① upon reaching position ② it directly jumps to position ③. Thus the phasor only covers an angle of 240° in the entire cycle instead of 360°



$$\Rightarrow \quad T' = \frac{240^\circ}{360^\circ} \times T = \frac{2}{3}T = \frac{2}{3}\cdot\frac{2\pi}{\omega} = \frac{4\pi}{3\omega}$$

where $\omega = \sqrt{\frac{k}{m}}$.

Methods of Generating SHM DE for mechanical systems:

(I) Force Method

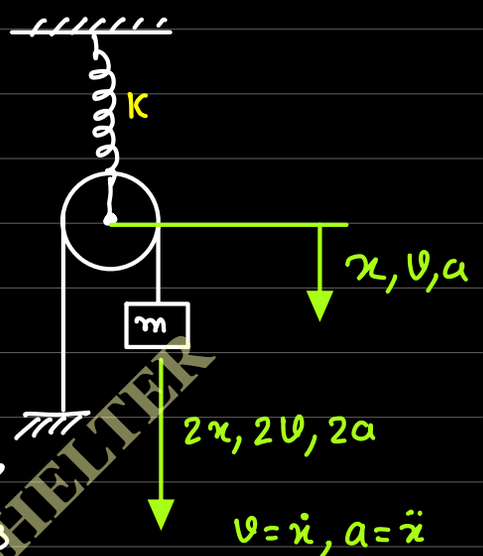
(II) Energy Method

(III) Torque Method

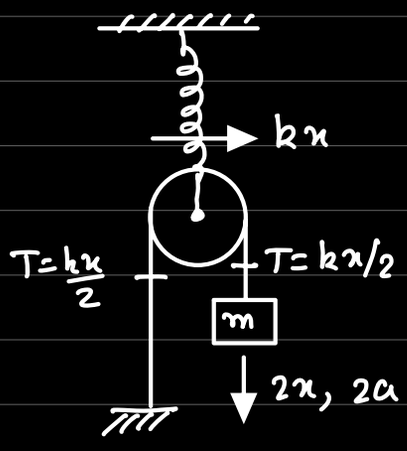
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Force Method: In this method we write $\vec{F} = m\vec{a}$ equations at a general coordinate representing configuration of a system (requires figuring out system constraint correctly).

Example: The system is released from rest with spring initially at natural configuration. Find the time period of oscillation of the system.



Sol: Let general elongation of spring be x,
$\Rightarrow$ block will descend by $2x$.
[To avoid any silly mistake, velocities and acc^n are also indicated multiplied by appropriate factors taking constraint in to consideration].



Now writing LOM equation:

$$mg - \frac{kx}{2} = m \times 2a = 2m\frac{d^2x}{dt^2}$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{4m}\left(x - \frac{2mg}{k}\right)$$

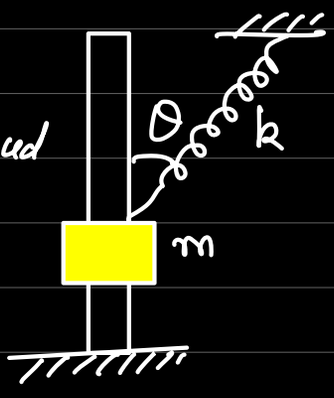
Comparing with SHM DE, $\omega = \sqrt{\frac{k}{4m}}$, $x_0 = \frac{2mg}{k}$

Note that since x was elongation of spring x_0 will be mean elongation in the spring and not mean position of the block. (x_0 represents the mean value of variable x).
Infact since displacement of block is $2x$, its mean position is $2x_0$ below the natural length position.

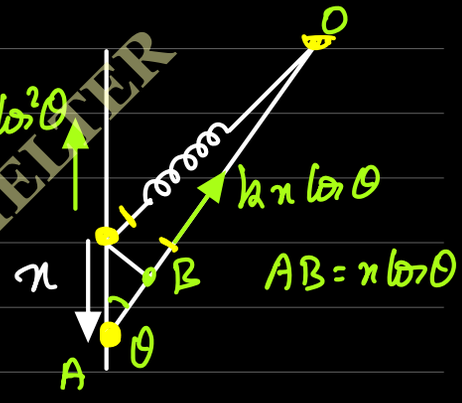
Now $T = 2\pi/\omega = 2\pi\sqrt{\frac{4m}{k}}$

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Effective Constant of a spring Collar Arrangement:



Let a collar be in equilibrium as shown in the figure. Now if the collar is displaced a little bit from its equilibrium position, the additional forces in the vertical direction are as shown in the figure.
For the small displacement x, the additional elongation in the spring is $x\cos\theta$ thus the additional spring force becomes $kx\cos\theta$ whose vertical component is simply $kx\cos^2\theta$.



Thus the arrangement behaves as if connected to a vertical spring of constant

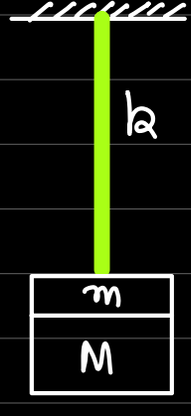
$$K_{eff} = k\cos^2\theta$$

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Difference between a Rubber Band and a Spring:

So long as a rubber band is in tension, it behaves like a spring but if the tension becomes zero it is just like a loose string.

Example: Two block of mass m and M are glued to each other as shown in the figure and attached to an elastic string (rubber band). Now the glue fails. Find max permissible value of M so that the subsequent motion remains simple harmonic.



Sol: Initial elongation: $x_0 = (M+m)g/k$.

New mean position: $x_0' = mg/k$. After the glue fails, the motion starts with zero velocity i.e extreme position therefore Amplitude:

$$A = x_0 - x_0' = Mg/k.$$

∴ If SHM continues, the m rises to a max height of $h_{max} = 2A = 2Mg/k$. And for SHM to take place, with this position, rubber band should still be in elongation (not loose)

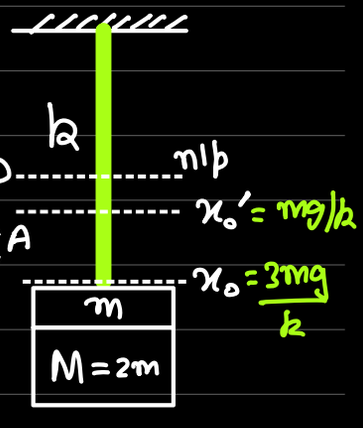
$$\therefore \quad h_{max} \leq (M+m)g/k$$

$$\Rightarrow 2Mg/k \leq (M+m)g/k \quad \Rightarrow \quad M \leq m \text{ Ans.}$$

→ Note: If $M > m$, motion will be like SHM until rubber band reaches natural length and after that it will be like a vertically thrown projectile.

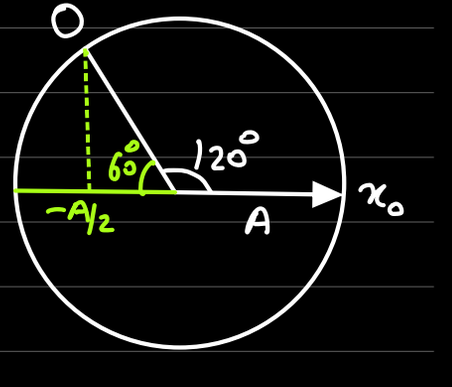
Q. In the previous example, if $M = 2m$, after how much time will the rubber band get loose. How will you find the cycle time period.

Sol: $A = \frac{(M+m)g}{k} - \frac{mg}{k} = 2mg/k.$



Thus the motion starts from $2mg/k$ below the mean position and the rubber band gets lose at a height mg/k i.e. $A/2$ above mean position. This can be easily visualised using a phasor diagram. From the phasor diagram, it is clear that required time $= \frac{120^\circ}{360^\circ} \times \frac{2\pi}{\omega}$

$$\Rightarrow \quad t_1 = 2\pi\sqrt{\frac{m}{k}} \times \frac{1}{3} = \frac{2\pi}{3}\sqrt{\frac{m}{k}} \text{ Ans.}$$



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Now speed of the block at this point is $v = \omega\sqrt{A^2 - x^2}$

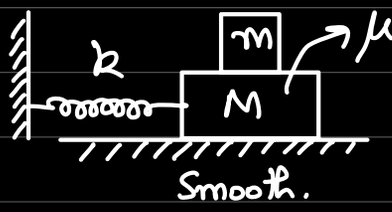
$$\Rightarrow \quad v = \omega\sqrt{A^2 - (A/2)^2} = \frac{\sqrt{3}}{2}A\omega$$

Now further time to reach the top $= \frac{v}{g} = \frac{\sqrt{3}A\omega}{2g}$

$$\Rightarrow \quad t_2 = \frac{\sqrt{3}}{g} \times \frac{2mg}{k} \times \sqrt{\frac{k}{m}} = \sqrt{\frac{3m}{k}}$$

Now since upward and downward motions are symmetric $\therefore$ total cycle time $T = 2(t_1 + t_2) = 2\sqrt{\frac{m}{k}}\left[\frac{2\pi}{3} + \sqrt{3}\right]$ Ans.

Q. For the shown spring mass system, find the maximum allowable amplitude if blocks do not slip relative to each other?



Sol: For no slipping acceleration of top block must not exceed μg.

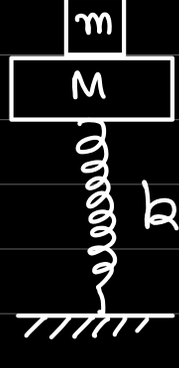
Now:

$$a_{max} = A\omega^2 \quad (\because x = A\sin\omega t)$$

$$\Rightarrow A\omega^2 < \mu g \Rightarrow A < \mu g/\omega^2 \Rightarrow A < \mu g/(k/M+m)$$

$$\Rightarrow A < \mu(M+m)g/k.$$

Q. Find the maximum allowable amplitude of oscillations so that m does not jump off M.

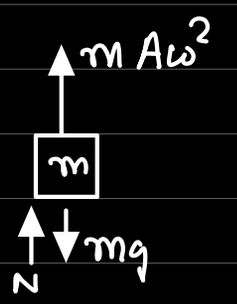


Sol: For no jumping mass upward pseudo force must not exceed mg.

$$\omega = \sqrt{k/M+m}$$

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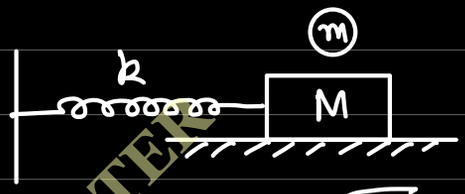
Now FBD of m at upper extreme position:



For $N>0$, $mA\omega^2 < mg \Rightarrow A < g/\omega^2$

$\Rightarrow A < (M+m)g/k$.

Q. A block of mass M is executing SHM of amplitude A as shown in the figure. Now a dough ball of mass m is dropped on M when it was at $A/2$ from mean position, Calculate the new amplitude.



$\omega = \sqrt{k/M}$

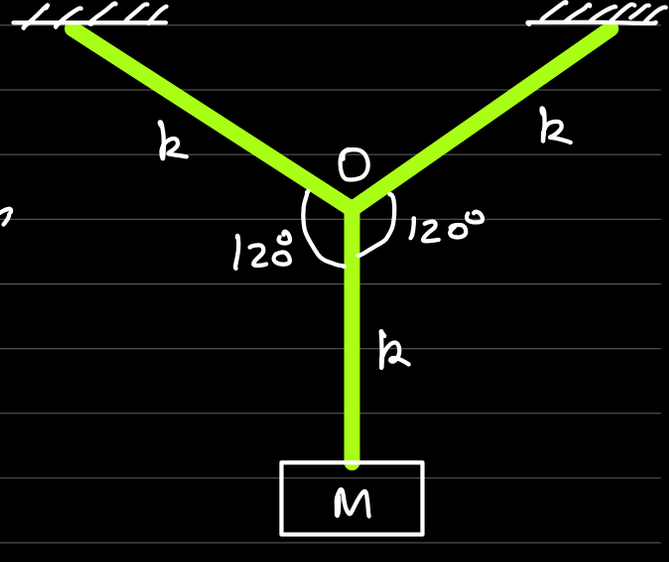
Sol: Initial energy $= E_0 = \frac{1}{2}kA^2$. Energy lost $= \frac{1}{2}\mu v_{rel}^2$

$$v_{rel} = \omega\sqrt{A^2-(A/2)^2} = \frac{\sqrt{3}}{2}A\omega$$

$$\Rightarrow \text{New Energy } E' = \frac{1}{2}kA^2 - \frac{1}{2}\left(\frac{Mm}{M+m}\right)\frac{3}{4}A^2\omega^2 = \frac{1}{2}kA'^2$$

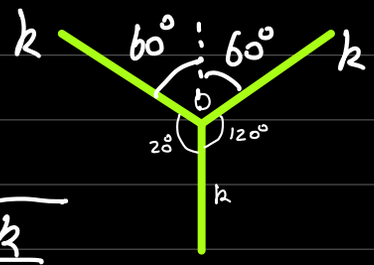
$$\Rightarrow \frac{1}{2}kA^2 - \frac{3}{8}k\frac{m}{M+m}A^2 = \frac{1}{2}kA'^2 \Rightarrow A' = A\sqrt{1-\frac{3m}{4(M+m)}}$$

Example: A block hangs in equilibrium as shown in the figure with help of three elastic strings as shown in the figure. Find the time period of small amplitude oscillation if block is slightly pulled down and released.



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Solution: **Method I:** Using spring collar analogy Each of the upper elastic strings behaves like a spring of effective constant $k\cos^2 60°$.



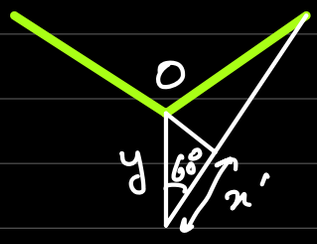
$$\Rightarrow K_{eff} = k \text{ Series } \left[\frac{k}{4} + \frac{k}{4}\right]$$

$$= k \times \left[\frac{1 \times 1/2}{1 + 1/2}\right] = k/3 \Rightarrow \omega = \sqrt{\frac{k}{3m}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{3m}{k}}.$$

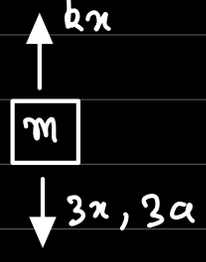
Method II: Let x be the additional elongation in the lower string. Now since O is a massless point, net force on it must be zero. Let x' be additional elongations in upper strings, then by Lami's theorem:

$$\frac{kx}{\sin 120°} = \frac{kx'}{\sin 120°} \Rightarrow x' = x.$$



Now due to x' elongation, O comes down by an amount, $y = x' \sec 60° = 2x$

$\Rightarrow$ Total descent of block $= y + x = 3x$
($\because$ lower string has also elongated by x).

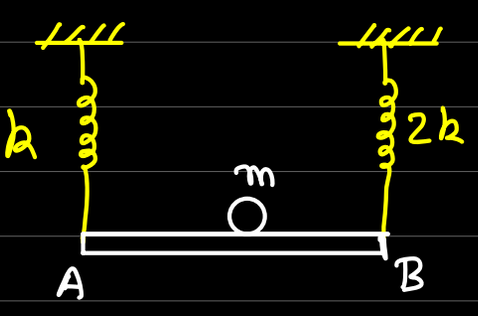


Now $mg - kx = m \times 3\frac{d^2x}{dt^2} \Rightarrow \omega = \sqrt{\frac{k}{3m}}$

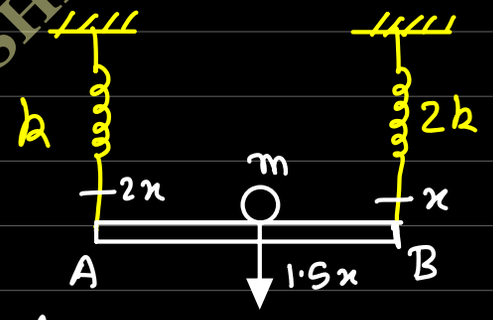
$$\Rightarrow T = 2\pi\sqrt{\frac{3m}{k}} \quad \text{Ans.}$$

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Example [Silly mistake Alert!]: AB is a massless rod with a point mass welded at its mid point as shown. The system is in equilibrium with the rod in horizontal position. Find the time period of small vertical oscillations of the point mass.



Sol.: ∵ the rod is massless, the net torque on it must be zero. Considering torques about its mid point it is obvious that elongations in spring must be in the inverse ratio of spring constants.
Also the descent of mid point of the rod must be average of descents of its end points (see figure) ∵ rod is a rigid body.



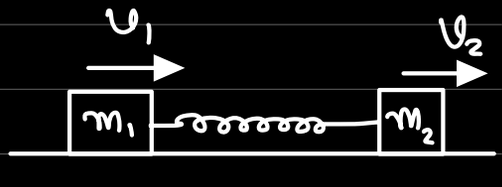
Now using $\vec{F}_{ext} = \sum m_i \vec{a}_i$ with additional forces:

$$-4kx = m \times 1.5 \frac{d^2x}{dt^2} \Rightarrow \omega = \sqrt{\frac{8k}{3m}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{3m}{8k}} \quad \text{Ans.}$$

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Two Connected blocks with spring on a smooth surface:



Let x be elongation of the spring, now considering FBD of block 2 from CM frame:

$$-kx\hat{i} = \frac{d}{dt}(\mu v_{21})\hat{i} = \mu\frac{d^2x}{dt^2}\hat{i}$$

$$\Rightarrow \boxed{T = 2\pi\sqrt{\frac{\mu}{k}}}$$

Example: Consider the system placed on a smooth horizontal surface with spring initially at natural length and m_2 given a speed u as shown.



(a) From CM frame find the amplitude of oscillation of m_2.

(b) Where is the mean position of m_2 as seen from CM frame.

(c) Draw the phasor diagram of m_2 @ $t=0$ and hence find its location wrt its mean position at a general time.

Sol. From CM frame, CM remains at rest thus individual displacements of the blocks must be in the inverse ratio of masses and in opposite directions. Also sum of their displacement magnitudes must be the elongation in the spring. Here since initial PE is zero and CM is an inertial frame, the initial position itself must be the mean position. (∵ Mean position is the position where net force is zero).

Let A_1 and A_2 be the amplitudes of oscillations of two blocks and A be the amplitude of length oscillations of the spring then:

$$A_1 + A_2 = A \quad (1) \quad (\because \text{CCOM})$$

also: $$\frac{1}{2} k A^2 = \frac{1}{2} \mu u^2 \quad (2)$$

Also $$m_1 A_1 = m_2 A_2 \quad (3)$$

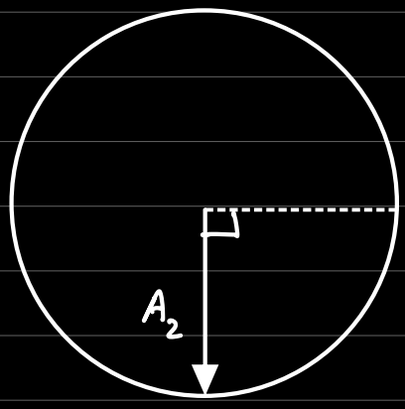
Now from (2), $$A = u\sqrt{\frac{\mu}{k}} = A_2\left[1 + \frac{m_2}{m_1}\right] \quad (\because (1) \& (3))$$

$$\Rightarrow A_2 = \frac{u m_1}{m_1 + m_2}\sqrt{\frac{\mu}{k}} \quad \text{Ans}$$

Now initial position is the standard mean position with rightward

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velocity. This gives us a phasor diagram as shown in the figure:



$\Rightarrow x_2 = A_2 \cos(\omega t - \frac{\pi}{2}) = A_2 \sin(\omega t)$ Ans.

where $\omega = \sqrt{\frac{k}{\mu}}$ and A_2 was found above.

Example: Consider the system placed on a smooth horizontal surface with spring initially at natural length and a force F applied on m_2 as shown.



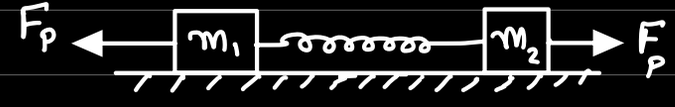
(a) From CM frame find the amplitude of oscillation of m_2.

(b) In CM frame where is mean position of m_2 wrt its initial position?

(c) Draw the phasor diagram of m_2 @ $t=0$ and hence find its location wrt its mean position at a general time. (Also find coordinate wrt its initial position)

Sol. $a_{cm} = \frac{F}{m_1+m_2} \Rightarrow$ Pseudo force on m_1:

$$F_p = \frac{m_1 F}{m_1+m_2}$$



Thus FBD from CM frame looks as shown in the figure (Recall that CM frame is a zero momentum frame $\therefore$ net force in CM frame on the system is zero).

Now from CM frame, the initial velocity of blocks is zero. $\therefore$ the initial position must be the extreme position. The mean position can be found by balancing the forces ($\because$ acc^n at mean position is zero).

Let A_1 and A_2 be the oscillation amplitudes in CM frame of individual blocks and A be the oscillation amplitude of

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length of the spring then:

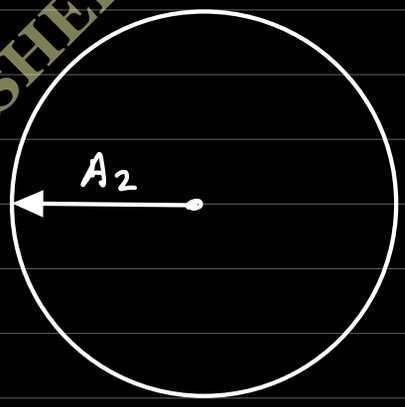
$$A = A_1 + A_2 \quad (1)$$

Also $m_1 A_1 = m_2 A_2$ ($\because$ CCOM in CM frame)
and by force balance:

$$F_p = kA \Rightarrow A = \frac{F_p}{k} = A_2\left[1 + \frac{m_2}{m_1}\right]$$

$$\Rightarrow A_2 = \frac{F_p}{k} \times \frac{m_1}{m_1 + m_2} = \frac{m_1^2}{(m_1 + m_2)^2} \cdot \frac{F}{k} \quad \text{Ans}$$

Now the initial position of m_2 is the negative extreme position ($\because$ velocity is zero and rightward motion is about to begin).



Thus coordinate of m_2 at a general time wrt its mean position is given as:

$$x = A_2 \cos(\omega t - \pi) = -A_2 \cos \omega t \quad \text{Ans.}$$

Now coordinate of mean pos. wrt initial pos^n is A_2 $\Rightarrow$ coordinate of m_2 wrt initial position:

$$x_2 = A_2 - A_2 \cos \omega t = A_2(1 - \cos \omega t) \quad \text{Ans.}$$

Here A_2 has been already found above and $\omega = \sqrt{\frac{k}{\mu}}$

Note: We could also find displacement of the block wrt ground by adding the displacement of CM i.e $\frac{1}{2} a_{cm} t^2$.

Torque method: This method is somewhat like force method, here we use $\tau = I\alpha$ to get a D.E. of the form:

$$I\frac{d^2\phi}{dt^2} = -\omega^2(\phi - \phi_0)$$ where ϕ is angular displacement of the body, ϕ_0 is the mean angular position and ω is the angular frequency of oscillation.

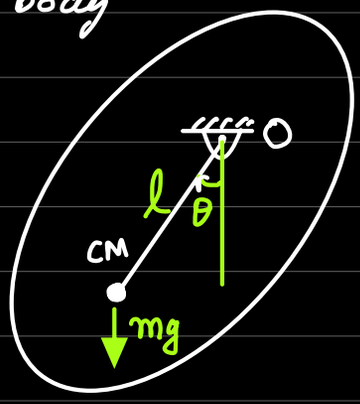
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[Note: $\omega \neq \frac{d\phi}{dt}$, $\frac{d\phi}{dt}$ represents angular velocity of the body which varies with time while ω is the angular velocity of the phasor].

The solution to the angular oscillation DE is given as:

$\phi - \phi_0 = A\sin(\omega t + \delta)$ where δ is the phase angle depending on initial ϕ and $d\phi/dt$, and A is the angular amplitude of oscillation.

PHYSICAL PENDULUM: Let an arbitrary rigid body be pivoted about a horizontal axis O with CM at a distance l from the pivot. For such a system the time period of oscillation is given as:



Small amplitudes Only → $$T = 2\pi\sqrt{\frac{I_0}{mgl}}$$

Where I_0 is the moment of inertia about O, m the mass of the rigid body. (This arrangement is known as physical pendulum).

Proof: Using $\tau_A = I_A\alpha$: $\quad mgl\sin\theta = -I_0\frac{d^2\theta}{dt^2}$

(Clockwise is taken as +ve θ) $\quad \Rightarrow mgl\theta \approx -I_0\frac{d^2\theta}{dt^2}$

($\because \theta$ is small).

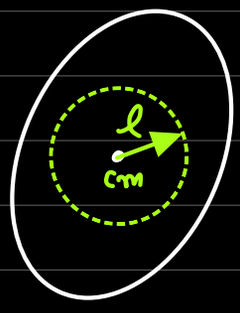
$$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{mgl}{I_0}\theta \quad \Rightarrow \omega = \sqrt{\frac{mgl}{I_0}}$$

$$\Rightarrow \boxed{T = 2\pi\sqrt{\frac{I_0}{mgl}}}$$

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Condition for minimum time period : Using ∥ axis theorem, $I_o = I_{cm} + ml^2 \Rightarrow T = 2\pi\sqrt{\frac{(I_{cm}+ml^2)}{mgl}}$

This means for all axes equidistant from CM, (see dotted circle) time period is same.

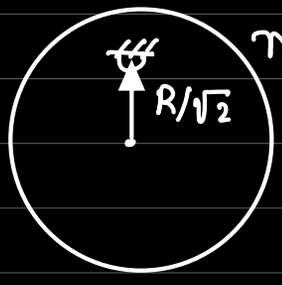


Also, $T^2 \propto \frac{I_{cm}}{ml} + l$, thus for minimum time period, $\frac{d}{dl}\left(\frac{I_{cm}}{ml} + l\right) = 0$

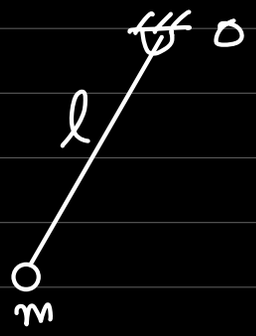
$\Rightarrow \frac{I_{cm}}{m}\left[\frac{-1}{l^2}\right] + 1 = 0$ or $l = \sqrt{\frac{I_{cm}}{m}}$ = radius of Gyration about CM.

Thus for minimum time period, the pivot must be at a distance equal to radius of Gyration about CM.

For example, for a disc least time period will occur when pivoted at a distance of $R/\sqrt{2}$ from CM as shown.

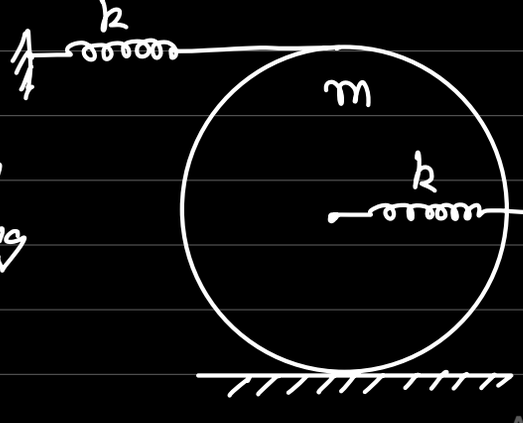


Simple Pendulum: For a simple pendulum, a point mass m is suspended from a stick of length l, thus $I_o = ml^2$, Putting in the formula for the physical pendulum:



$$\boxed{T = 2\pi\sqrt{\frac{l}{g}}}$$

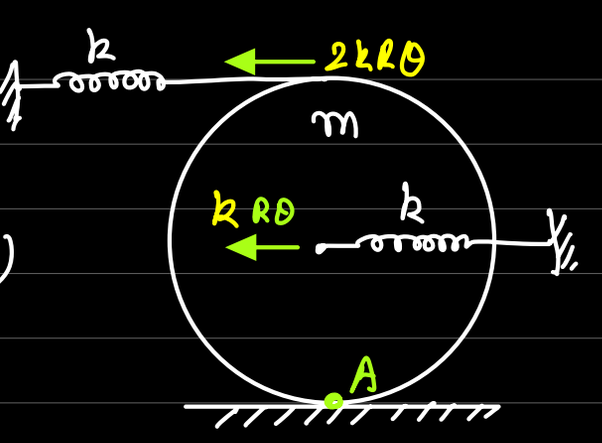
Example: Find the time period of oscillation of the small oscillations of the disc shown in the figure. Assume no slipping condition.



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Solution: Let the disc turn through a small angle θ. Now the displacement of the centre must be $R\theta$

($\because v = \omega R, \Rightarrow \frac{dx}{dt} = R\frac{d\theta}{dt} \Rightarrow x = R\theta$)



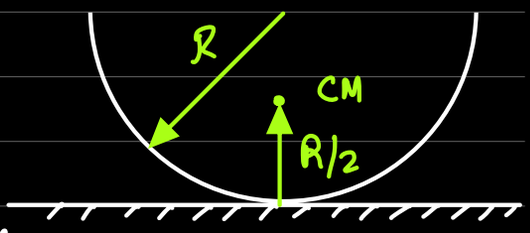
Now taking torques about A:

$$kR\theta \times R + (k \times 2R\theta) \times 2R = -\frac{3}{2} mR^2 \frac{d^2\theta}{dt^2}$$

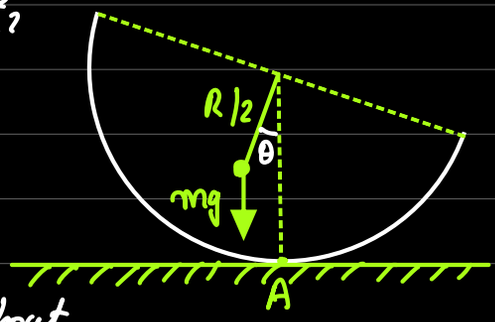
$$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{10k}{3m}\theta \quad \Rightarrow \omega = \sqrt{\frac{10k}{3m}}, \quad T = 2\pi\sqrt{\frac{3m}{10k}} \text{ Ans}$$

<u>Example:</u> A hemispherical Bowl of mass m and radius R rests on a rough horizontal surface, find the time period of oscillations.

Solution: Consider FBD of the bowl after tilting through an angle θ.



Using torque equation: $mg\frac{R}{2}\sin\theta = -I_A\frac{d^2\theta}{dt^2}$



$$I_A = \frac{2}{3}mR^2 - m\left(\frac{R}{2}\right)^2 + m(R/2)^2$$

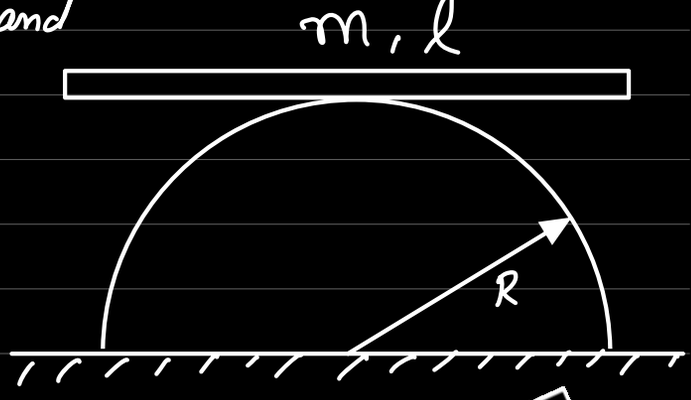
(θ is very small and angular velocity is also very small $\therefore$ torque of pseudo force about A is ignored)

$$\Rightarrow \frac{d^2\theta}{dt^2} = \frac{-3g}{4R}\theta \qquad (\because \sin\theta \propto \theta)$$

$$\Rightarrow T = 2\pi\sqrt{\frac{4R}{3g}} \text{ Ans.}$$

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Example: A thin rod of mass m and length ℓ rests on a sufficiently rough semi-cylinder as shown. Find the time period of oscillations.

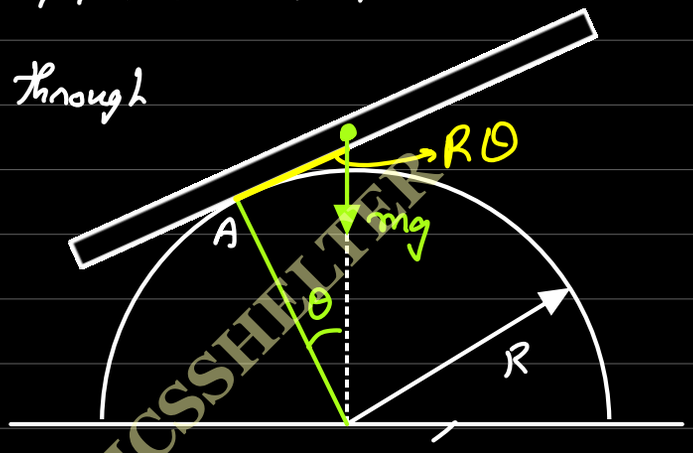


Solution: Consider a small angle θ through which rod is rotated. Taking torque about A:

$$mgR\sin\theta = -\frac{m\ell^2}{12}\frac{d^2\theta}{dt^2}$$

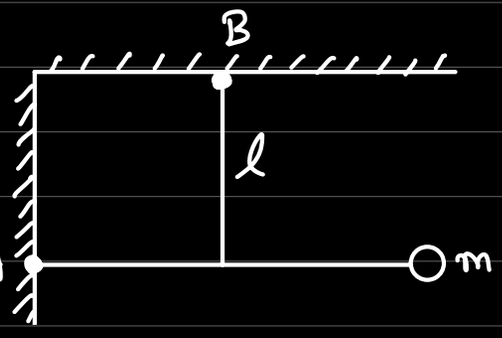
$$\Rightarrow \omega = \sqrt{\frac{12gR}{\ell^2}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{\ell^2}{12gR}} \text{ ans.}$$

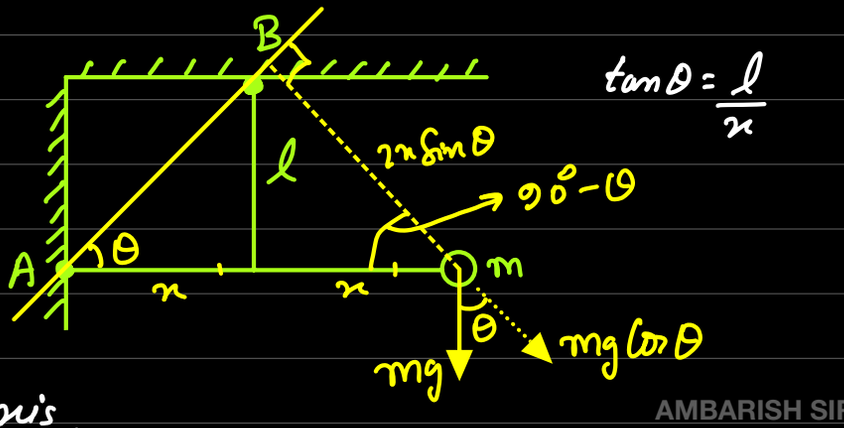


→ Note: ∵ θ is small change in moment of inertia and pseudo force torque were ignored (Second order terms).

Example: A weightless rigid rod with a load at the end is hinged at point A (ball and socket joint). The rod is kept in a horizontal position by a thread connected to its mid point. Find the time period of small amplitude oscillations if m is given a slight push inwards.



Solution: It can be readily seen that the line AB acts like permanent axis of the rod and only $mg\cos\theta$ produces a torque about this axis.



$$\tan\theta = \frac{\ell}{x}$$

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($\because$ $mg\sin\theta$ is $\parallel$ to the axis). Thus this problem becomes just like a physical pendulum with $g_{eff} = g\cos\theta$,

$l_{eff} = 2x\sin\theta$ and $I_0 = m(4x^2\sin^2\theta)$.

Now using $T = 2\pi\sqrt{\frac{I}{mgl}}$, $T = 2\pi\sqrt{\frac{4mx^2\sin^2\theta}{mg\cos\theta \times 2x\sin\theta}}$

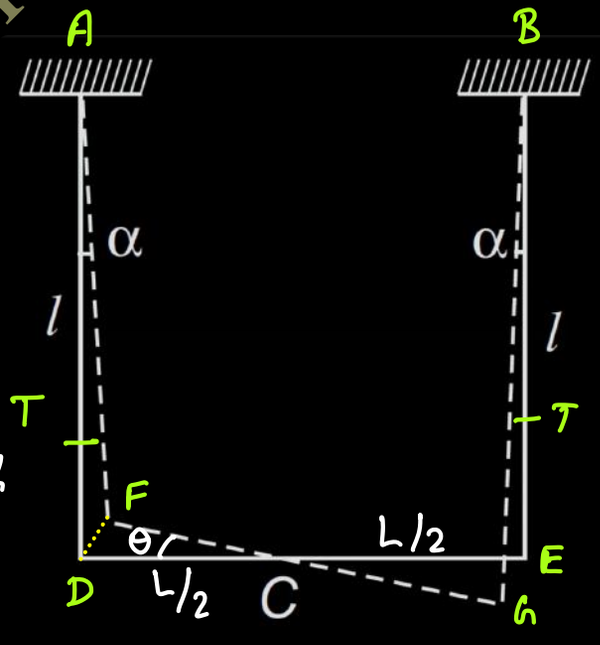
$\Rightarrow T = 2\pi\sqrt{\frac{2x\tan\theta}{g}}$ but $\tan\theta = l/x$

$\Rightarrow T = 2\pi\sqrt{\frac{2l}{g}}$ Ans.

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Example: Bifilar Pendulum:

A uniform rod of mass m is suspended by two identical threads of length l each. The rod was turned through a small angle θ about the vertical axis passing through its mid point C. Find
(a) Time period of small oscillations
(b) Energy of oscillation.



Sol: Since the oscillations are of small amplitude both tensions must be $mg/2$ each. (Small accelerations). The left thread lies in the plane ADF whose component along FD will produce a torque about the C axis. (Similar contribution will come from right thread also).

Now making torque equation about C axis:

$$\left(\frac{mg}{2}\sin\alpha \times \frac{L}{2}\right) \times 2 = -\frac{mL^2}{12}\frac{d^2\theta}{dt^2} \quad (1)$$

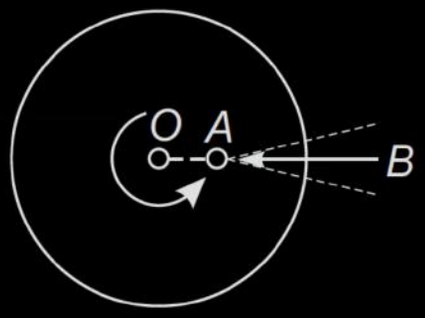
also $FD = l\alpha = \frac{L}{2}\theta \Rightarrow \sin\alpha \approx \alpha = \frac{L}{2l}\theta$

$$\Rightarrow \frac{mgL^2}{4l}\theta = -\frac{mL^2}{12}\frac{d^2\theta}{dt^2} \Rightarrow \frac{d^2\theta}{dt^2} = \frac{3g}{l}$$

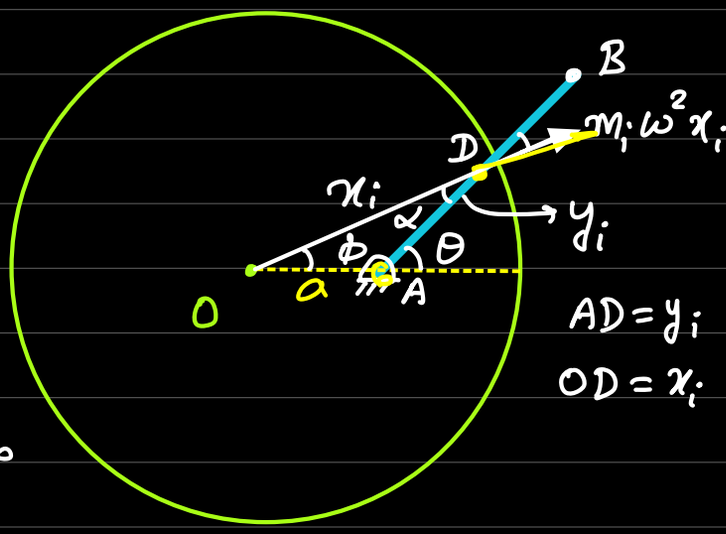
$$\Rightarrow T = 2\pi\sqrt{\frac{l}{3g}} \text{ Ans.}$$

Example: (Oscillations due to centrifugal force)

A smooth horizontal disc rotates about the vertical axis O (Fig. shown) with a constant angular velocity ω. A thin uniform rod AB of length l performs small oscillations about the vertical axis A fixed to the disc at a distance a from the axis of the disc. Find the frequency ω_0 of these oscillations.



Solution. From the rotating frame of the disc, ∵ velocity of each particle of rod is ⊥ to the rod, and coriolis force $= -2m(\vec{\omega} \times \vec{v}_{rot})$, the coriolis forces will be along the rod and ∴ produce no torque about the pivot A. The only torque about pivot is that of centrifugal force

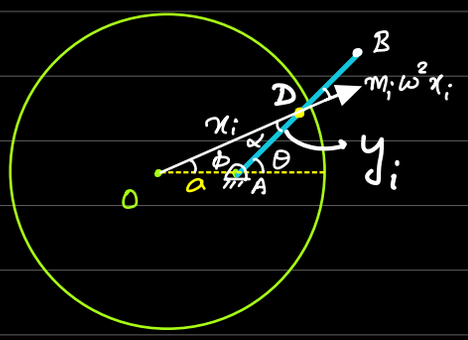


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$$\vec{\tau}_i = -m_i\omega^2 \vec{OD} \times \vec{AD}$$

$$= -m_i\omega^2(\vec{OA} + \vec{AD}) \times \vec{AD}$$

$$= -m_i\omega^2(\vec{OA} \times \vec{AD})$$



$$\Rightarrow |\vec{\tau}_i| = m_i\omega^2 a y_i \sin\theta \Rightarrow \tau = \omega^2 a \sin\theta \sum m_i y_i$$

$$\Rightarrow \tau = m\omega^2 a \sin\theta \frac{l}{2} \approx \frac{m\omega^2 a l\theta}{2} \quad (\because \theta \text{ is very small})$$

Now using $\tau = I\alpha$, $\quad \frac{m\omega^2 a l\theta}{2} = -\frac{ml^2}{3}\frac{d^2\theta}{dt^2}$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{3a\omega^2}{2l}\theta \Rightarrow \text{required angular frequency:}$$

$$\omega_0 = \sqrt{\frac{3a\omega^2}{2l}} \quad \text{Ans.}$$

[Note: We can't apply centrifugal force here at CM ∵ centrifugal field is not uniform]

ENERGY METHOD (Very useful in complicated set ups):

In this method we use the following steps:

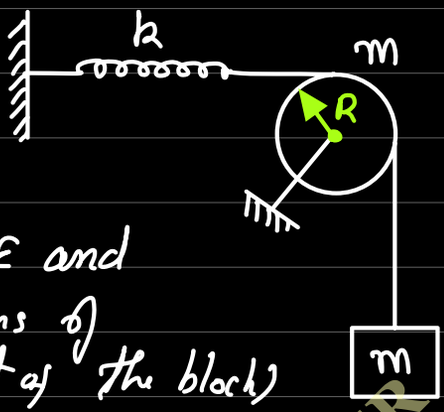
① Write $KE(x) + PE(x) = \text{Constant}$ (I)
or $KE(\theta) + PE(\theta) = \text{Constant}$ where θ or x is a general variable specifying configuration of the system.

② Differentiate the equation of step ① wrt time and cancel a suitable factor throughout (generally velocity or angular velocity is cancelled)

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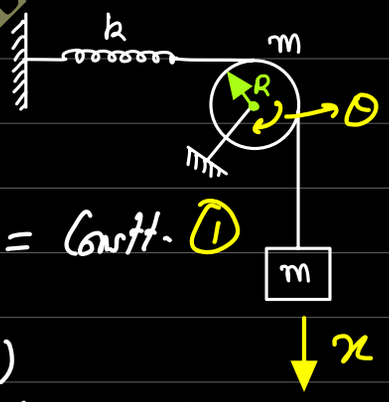
③ Rearrange the equation obtained in step ② in to standard SHM DE.

Example: Find the time period of oscillation of the system shown in the figure.



Solution: Here we shall express the KE and PE of the system either in terms of linear variable (x i.e displacement of the block) or in terms of angular variable (θ i.e rotation of pulley).

Now let block come down by a distance x and pulley rotate through an angle θ then:



$$\frac{1}{2}kx^2 + \frac{1}{2}\times\frac{1}{2}mR^2\omega^2 + \frac{1}{2}mv^2 - mgx = \text{Constt.} \quad (1)$$

Now we know that $\omega R = v$ $(\because R\theta = x)$

Now before differentiating wrt we need to make the entire equation either using ω & θ or x & v.

$$\Rightarrow \frac{1}{2}kx^2 + \frac{1}{4}mv^2 + \frac{1}{2}mv^2 - mgx = C$$

$$\Rightarrow \frac{1}{2}kx^2 + \frac{3}{4}mv^2 - mgx = \text{Constt.} \leftarrow \text{Now Differentiate:}$$

$$\Rightarrow kx\dot{x} + \frac{3}{2}mv\frac{d^2x}{dt^2} - mgv = 0 \leftarrow \text{Now cancel } v$$

$$\Rightarrow \frac{3}{2}m\frac{d^2x}{dt^2} = -kx + mg \Rightarrow \frac{d^2x}{dt^2} = \frac{-2k}{3m}\left(x - \frac{mg}{k}\right)$$

$$\Rightarrow x_0 = \frac{mg}{k} \text{ and } T = 2\pi\sqrt{\frac{3m}{2k}} \text{ Ans.}$$

Note: Alternatively we could put $x = R\theta$ and $v = R\omega$ in (1) and cancel θ throughout after differentiating. In that case,

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before differentiating we would get:

$$\frac{1}{2}k(R\theta)^2 + \frac{3}{4}m(R\omega)^2 - mgR\theta = C$$

Now Diff. wrt time:

$$kR^2\theta\dot{\theta} + \frac{3}{2}mR^2\dot{\theta}\frac{d^2\theta}{dt^2} - mgR\dot{\theta} = 0$$

Now we can divide by $\dot{\theta}$ and then rearrange to get the same time period.

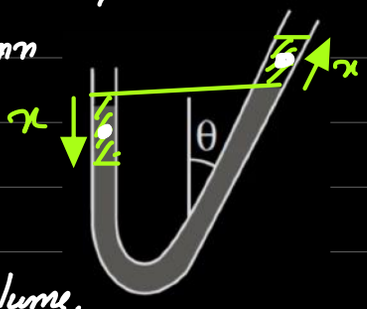
Example:

Determine the period of oscillations of mercury of mass $m = 200$ g poured into a bent tube (Fig. shown) whose right arm forms an angle $\theta = 30°$ with the vertical. The cross-sectional area of the tube is $S = 0.50$ cm^2. The viscosity of mercury is to be neglected.



Fig. 4.5

Solution: let level of liquid fall by a distance x in left limb, then length of the liquid in right column will increase by same amount due to volume conservation. Now for calculating change in PE, we can think that as if left shaded volume is transferred to the right shaded volume.



Now we need to find the change in PE due to this transfer. We need to see change in height of CM of this shaded portion.

$\therefore$ PE$(x) = \rho S x \, x\left(\frac{x}{2} + \frac{x}{2}\cos\theta\right)g$ ($\because$ initial CM was $\frac{x}{2}$ below the free surface and final CM is $\frac{x}{2}\cos\theta$ above the free surface.

$$\therefore \frac{\rho S x^2 g(1+\cos\theta)}{2} + \frac{1}{2}mv^2 = C$$

$$\Rightarrow \rho S x v g(1+\cos\theta) + mv\frac{d^2x}{dt^2} = 0$$

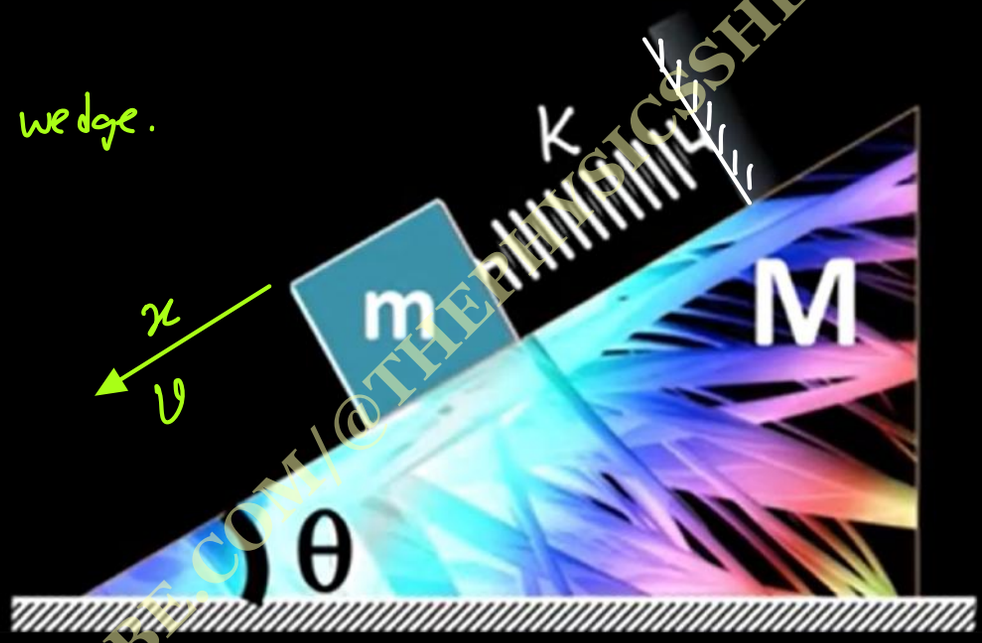
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$$\Rightarrow \quad \frac{d^2x}{dt^2} = -\frac{\rho S x g (1+\cos\theta)}{m} \quad \Rightarrow \quad T = 2\pi\sqrt{\frac{m}{\rho S g (1+\cos\theta)}} \text{ Ans.}$$

Example:

SHM of a Spring Block Wedge System

A Block of mass m rests on a frictionless wedge of mass M being connected to a spring of constant K as shown in the figure. The ground is frictionless. Calculate the time period of small oscillations.



x, v : Relative to wedge.

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Solution: Recall that KE of a two body system can be written as $KE = \frac{1}{2} M_0 v_{cm}^2 + \frac{1}{2} \mu v_{rel}^2$ where $M_0 = M+m$.

Thus by energy conservation:

$$\frac{1}{2}(M+m)x\left(\frac{mv\sin\theta}{M+m}\right)^2 + \frac{1}{2}\left(\frac{Mm}{M+m}\right)v^2 + \frac{1}{2}kx^2 - mgx\sin\theta = \text{Constt.}$$

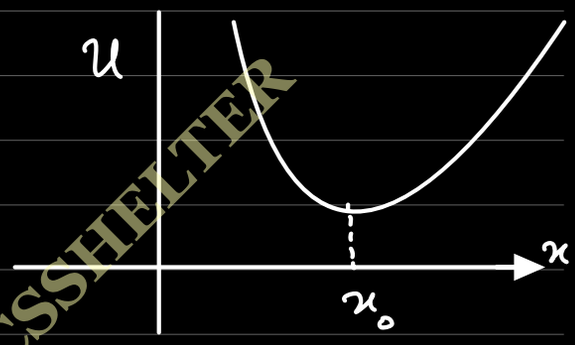
$$\Rightarrow \frac{mv^2}{2(M+m)}\left[m\sin^2\theta + M\right] + \frac{1}{2}kx^2 - mgx\sin\theta = \text{Constt.}$$

$$\Rightarrow \quad \frac{mv}{M+m}\left[M+m\sin^2\theta\right]\frac{d^2x}{dt^2} + kxv - mgv\sin\theta = 0$$

Now divide by v and rearrange, we finally get:

$$T = 2\pi\sqrt{\frac{m(M+m\sin^2\theta)}{(M+m)k}} \quad \text{Ans.}$$

Oscillations in a 1D Potential Energy Field (Translating Particle)



We can easily show that if a particle of mass m is in a stable equilibrium position at $x = x_0$ then upon small perturbation, its time period of oscillation is given as:

$$\boxed{T = 2\pi\sqrt{\frac{m}{U''(x)}}}$$

Proof: Recall that $F(x) = -U'(x)$

Also that $F(x+\Delta x) = F(x) + F'(x)\Delta x$

Now let x_0 be the equilibrium position and we move a small distance x ahead of it then:

$$F(x_0 + x) = F(x_0) + F'(x_0) \times x$$

But $F(x_0) = 0$ ($\because$ equilibrium), Also $F'(x_0) = -U''(x_0)$

$\Rightarrow$ Force on particle: $F = -U''(x_0)x = m\frac{d^2x}{dt^2}$

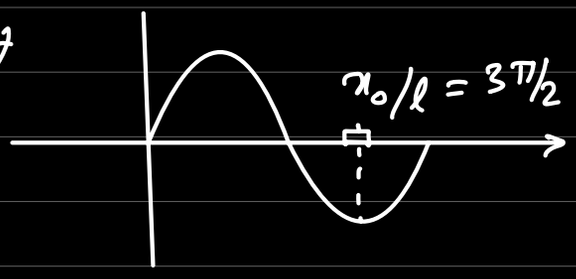
$$\Rightarrow \quad T = 2\pi\sqrt{\frac{m}{U''(x_0)}}$$

Example: A particle is in a stable equilibrium in a P.E. field given as $U = U_0\sin\left(\frac{x}{l}\right)$ where U_0 and l are appropriate constants.

Find time period of small oscillations if mass of particle is m.

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Sol: For stable equilibrium possible must be at a minima (such as $x_0 = \frac{3\pi}{2} l$)



Now $U' = \frac{U_0}{l} \cos(x/l)$

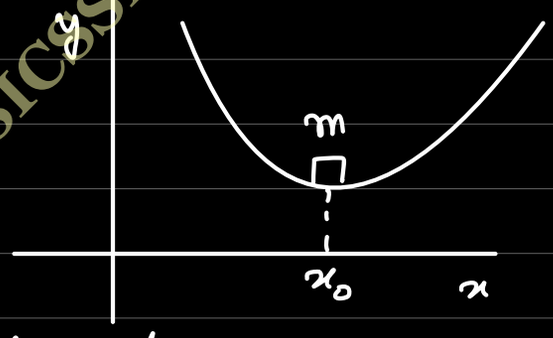
$U'' = -\frac{U_0}{l^2} \sin(x/l)$ @ $x = \frac{3\pi l}{2}$, $U'' = U_0/l^2$

$\Rightarrow T = 2\pi \sqrt{\frac{m}{(U_0/l^2)}} = 2\pi l \sqrt{\frac{m}{U_0}}$ Ans.

Time period in valley of a curve in a vertical plane: (Sliding mass)

Here $U(x) = mgy \Rightarrow U'' = mgy''$

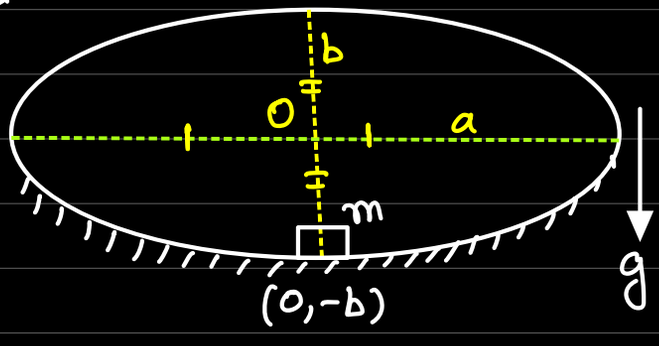
$\Rightarrow T = 2\pi \sqrt{\frac{m}{mgy''}} = 2\pi \sqrt{\frac{R(x_0)}{g}}$



Recally that $\frac{1}{y''}$ represents radius of curvature at peak on valley.

$\because R = (1+y'^2)^{3/2}/y'')$

Example: A block lies in valley of a smooth vertical elliptical track as shown in the figure. Find the time period of oscillation if the ellipse is given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



Sol: We need $y''(0)$ for the lower half of the ellipse for finding radius of curvature.

Now $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{2x}{a^2} + \frac{2yy'}{b^2} = 0$

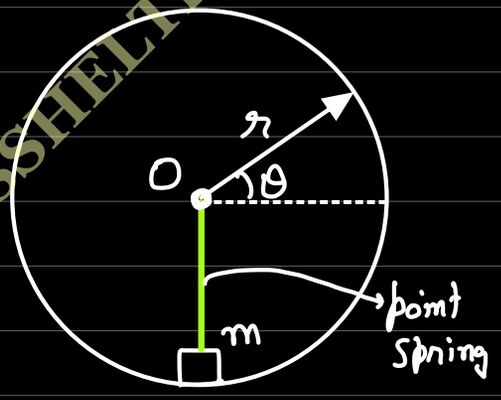
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$$\Rightarrow \quad \frac{2}{a^2} + \frac{2y'^2}{b^2} + \frac{2yy''}{b^2} = 0 \cdot \quad \text{For our point, } y' = 0,\ y = -b$$

$$\Rightarrow \quad \frac{2}{a^2} - \frac{2y''}{b} = 0 \quad \Rightarrow \quad y'' = \frac{b}{a^2} \Rightarrow R = a^2/b$$

$$\Rightarrow \quad T = 2\pi\sqrt{\frac{a^2}{bg}} \quad \text{Ans.}$$

Radial Oscillations in a Central Field:



Consider a particle undergoing motion in a central field $U(r)$. Now since the field is a central one, angular momentum of the particle must remain conserved after any disturbance.

By Energy conservation:

$$U(r) + \frac{1}{2}m\dot{r}^2 + \frac{1}{2}m(r\dot{\theta})^2 = \text{Const}$$

Also $L = mr^2\dot{\theta}$, $\Rightarrow \dot{\theta} = \frac{L}{mr^2}$

$$\Rightarrow \quad U(r) + \frac{L^2}{2mr^2} + \frac{1}{2}m\dot{r}^2 = \text{Constt.}$$

Now with $\dot{\theta}$ gone this becomes like a 1D problem, we can define an effective PE function as :

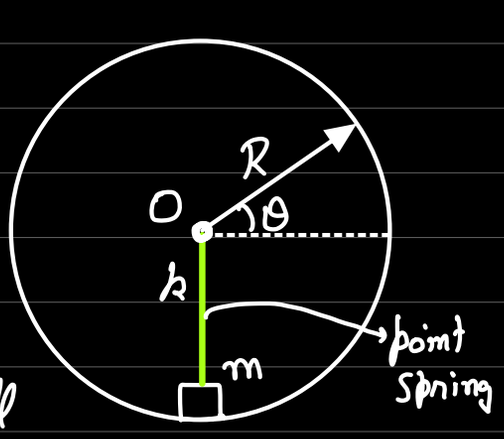
$$E(r) = U(r) + \frac{L^2}{2mr^2}, \quad \text{And problem now looks like}$$

same as particle in 1D potential field. $\therefore$ For radial oscillations, $\boxed{T = 2\pi\sqrt{\frac{m}{E''(r_0)}}}$

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Example: A particle is undergoing circular motion on a smooth horizontal table as shown (being connected to a point spring of constant k).

Find the time period of radial oscillations, if the particle is given a small radial kick.



Sol. $\frac{1}{2}kr^2 + \frac{L^2}{2mr^2} + \frac{1}{2}m\dot{r}^2 = \text{Const.}$ ①

Put $E = \frac{1}{2}kr^2 + \frac{L^2}{2mr^2} \Rightarrow E'' = k - \frac{d}{dr}\left[\frac{L^2}{mr^3}\right]$

$E'' = k + \frac{3L^2}{mr^4}$, Also, $L = mR^2\omega$ ② and $kR = m\omega^2 R$ ③

$\Rightarrow E''(R) = k + \frac{3L^2}{mR^4} = k + \frac{3}{mR^4}(m^2R^4\omega^2) = k + 3m\omega^2 = 4k$

$\Rightarrow T = 2\pi\sqrt{\frac{m}{E''(R)}} = 2\pi\sqrt{\frac{m}{4k}}$ Ans.

Amplitude Dependence of Time period if $U \propto x^n$:

Consider a particle connected to a point spring from origin with amplitude A and $U = Cx^n$

We shall try to find time taken to reach the distance A from origin.



$$\frac{1}{2}mv^2 = C(A^n - x^n)$$

$\Rightarrow v = k\sqrt{A^n - x^n} = \frac{dx}{dt}$ where k is some constant.

$\Rightarrow \int_0^A \frac{dx}{k\sqrt{A^n - x^n}} = \int_0^{T/4} dt$ Now put $\frac{x}{A} = y \Rightarrow dx = A\,dy$

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$$\Rightarrow \frac{T}{4} = \frac{1}{k}\int_0^1 \frac{A\,dy}{\sqrt{A^n - A^n y^n}} = \frac{A^{1-n/2}}{k}\int_0^1 \frac{dy}{\sqrt{1-y^n}}$$

Now even though we don't know the integral part, but we know it is some constant.

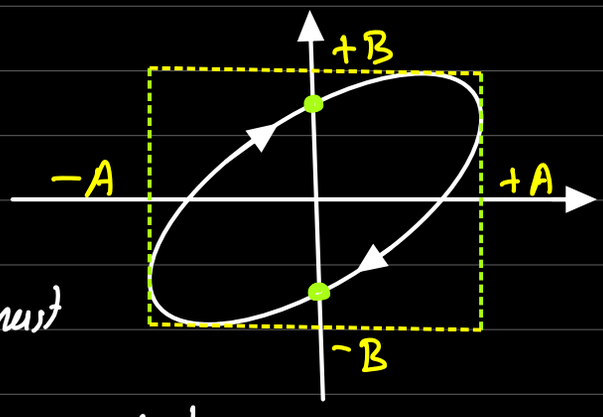
$$\therefore \quad \boxed{T \propto A^{1-n/2}}$$

It can be readily seen that for $n=2$ T is independent of A as expected in SHM.

Superposition of Mutually perpendicular SHM:

Consider a particle whose x and y coordinates are simultaneously varying Simple harmonically with same angular frequency.

i.e $x = A\sin(\omega t)$ and $y = B\sin(\omega t + \phi)$, By suitably eliminating t and using standard theory of coordinate geometry it can be shown that, the locus of particle is an ellipse.



Let us find the condition of ϕ for the particle to travel clockwise.

$\therefore$ While crossing +ve y axis, $\frac{dx}{dt}$ must be positive.

Now at $t=0$, $x=0$, $y = B\sin\phi$ and $\frac{dx}{dt} = A\omega\cos(\omega t)$

$\therefore \frac{dx}{dt} > 0$, $y>0$ for clockwise traversal $\Rightarrow B\sin\phi > 0$

$\Rightarrow \phi \in (0, \pi)$ Thus:

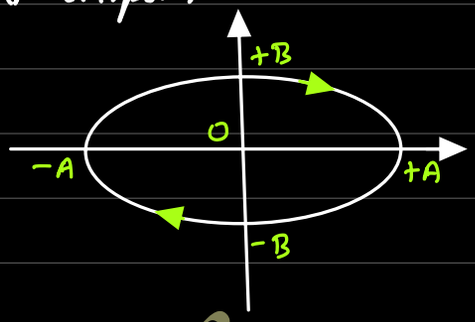
If $\phi \in (0,\pi) \Rightarrow$ Clockwise Ellipse, $\phi \in (0,-\pi) \Rightarrow$ Anticlockwise Ellipse

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Some special cases :

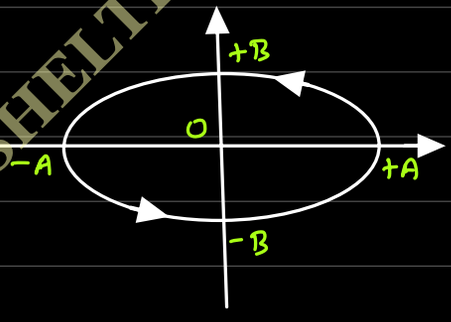
(I) $x = A\sin(\omega t)$, $y = B\sin(\omega t + \pi/2) = B\cos(\omega t)$

$\Rightarrow \quad \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1 \Rightarrow$ clockwise standard Ellipse.



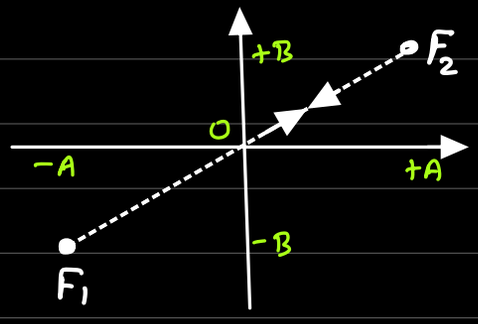
(II) $x = A\sin\omega t$, $y = B\sin(\omega t - \pi/2)$

$\Rightarrow \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1 \Rightarrow$ Standard ACW ellipse



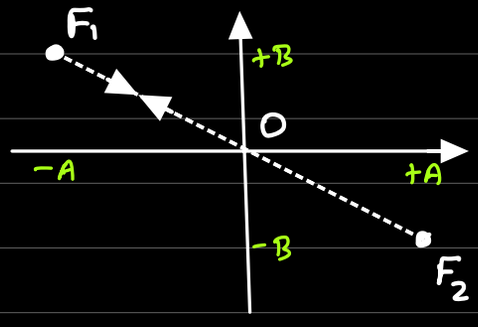
(III) $x = A\sin\omega t$, $y = B\sin(\omega t) \Rightarrow y = \frac{B}{A}x$

$\Rightarrow$ Straight line with positive slope.
(Degenerate ellipse, foci at ends)



(IV) $x = A\sin(\omega t)$, $y = B\sin(\omega t \pm \pi) \Rightarrow y = -\frac{B}{A}x$

$\Rightarrow$ Straight line with −ve slope



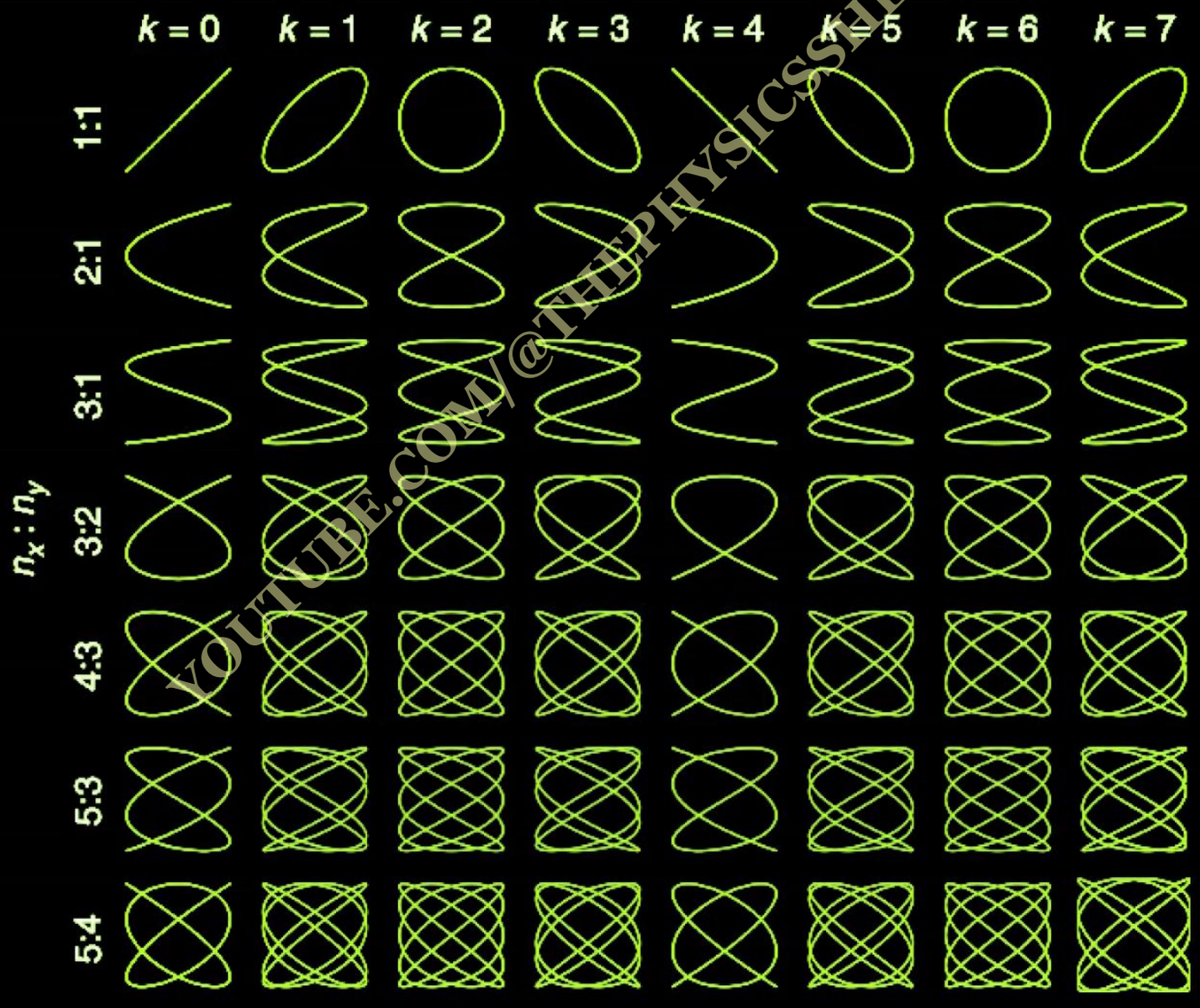
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Lissajous Figures: When x and y coordinates of a particle vary simple harmonically such that the ratio of angular frequencies in two directions is a rational number, the locus of the particle in such a case is called a lissajous figure. For a general 2D lissajous figure,

$$x = A\cos(m\omega t + \phi), \quad y = B\cos(n\omega t + \delta)$$

where m and n are integers.

Examples: $x = A_x\cos(n_x\omega t)$, $y = A_y\cos(n_y\omega t + k\pi/4)$

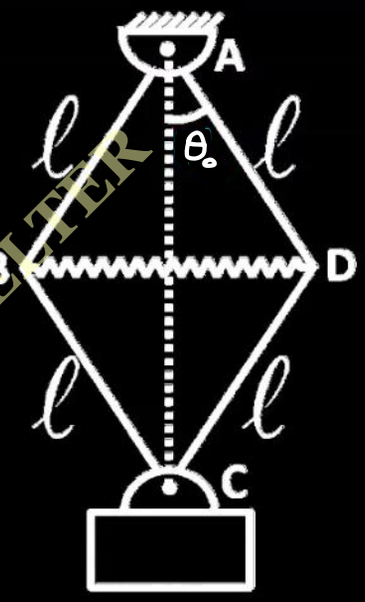


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Exercise: Try to find out x to y time period ratio by counting left right and up down cycles in complete curve traversal.

EXAMPLE 0 : KROTOV 1.96:

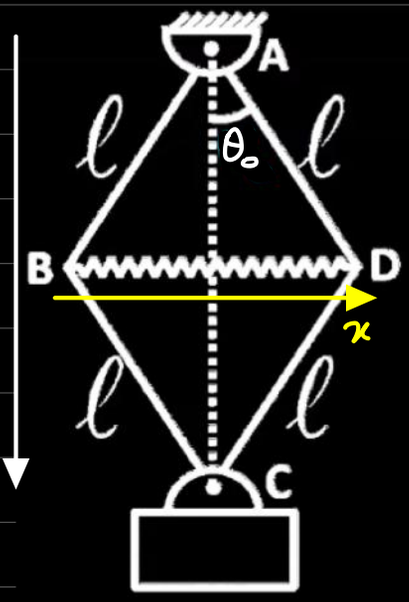
Four weightless rods of length l each are connected by hinged joints and form a rhomb (See Figure). The hinge A is fixed, and a load is suspended to a hinge C. Hinges D and B are connected by a weightless spring of natural length $1.5l$. In equilibrium, the rods form angles $\theta_0 = 30°$ with the vertical. Determine the period T of small oscillations of the load.



Mass of the block and spring constant are not known.

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<u>Solution:</u> Just as in terms of linear displacement variable, $T = 2\pi\sqrt{\frac{m}{U''(x_0)}}$ similarly in terms of angular variables $T = 2\pi\sqrt{\frac{I}{U''(\theta_0)}}$ where I represents some kind of term with dimensions of moment of inertia such that KE can be written as $KE = \frac{1}{2} I \dot{\theta}^2$.



For our case, $U = -2mg\ell\cos\theta + \frac{1}{2}k(1.5\ell - 2\ell\sin\theta)^2$

$\Rightarrow U' = 2mg\ell\sin\theta + k\ell^2(1.5 - 2\sin\theta)\cdot(-2\cos\theta)$

$\Rightarrow U'' = 2mg\ell\cos\theta + k\ell^2\frac{d}{d\theta}(-3\cos\theta + 2\sin 2\theta) = 2mg\ell\cos\theta + k\ell^2(3\sin\theta + 4\cos 2\theta)$

$= \sqrt{3}mg\ell + \frac{7}{2}k\ell^2 \quad (\because \theta_0 = 30°)$

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Now $KE = \frac{1}{2}m\dot{y}^2 = \frac{1}{2}m\cdot\left[\frac{d}{dt}(2l\cos\theta)\right]^2 = 2ml^2\sin^2\theta\,\dot{\theta}^2 = \frac{1}{2}I\dot{\theta}^2$

$\Rightarrow I = 4ml^2\sin^2\theta = ml^2 \quad (\because \theta_0 = 30^\circ)$

$$\Rightarrow T = 2\pi\sqrt{\frac{ml^2}{\sqrt{3}mgl + \frac{7}{2}kl^2}} = 2\pi\sqrt{\frac{ml}{\sqrt{3}mg + \frac{7}{2}kl}}$$

Now at equilibrium, $2T\cos 30^\circ = mg$ and $2T\sin 30^\circ = kx0.5l$

$\Rightarrow \tan 30^\circ = \frac{k}{2mg} = \frac{1}{\sqrt{3}} \Rightarrow mg = \frac{k\sqrt{3}}{2}l$

$$\Rightarrow T = 2\pi\sqrt{\frac{kl^2\sqrt{3}/2g}{\frac{3kl}{2} + \frac{7kl}{2}}} = 2\pi\sqrt{\frac{\sqrt{3}l}{10g}} \quad \text{Ans.}$$

METHOD 2: Force on load is $F = 2T\cos\theta$ while that on spring is $F_s = 2T\sin\theta$, thus $F = F_s\cot\theta = k[1.5l - 2l\sin\theta]\cot\theta$

$\Rightarrow F = 1.5kl\cot\theta - 2kl\cos\theta$

$\Rightarrow$ On small change in θ, $\Delta F = [-1.5kl\,\mathrm{cosec}^2\theta + 2kl\sin\theta]\,\Delta\theta$

for $\theta = 30^\circ$, $\Delta F = -5kl\Delta\theta$ also $y = 2l\cos\theta \Rightarrow \Delta y = -2l\sin\theta\Delta\theta$

$= -l\Delta\theta$

Thus $\Delta F = 5k\Delta y \Rightarrow$ effective spring constant $= 5k$

$\Rightarrow T = 2\pi\sqrt{\frac{m}{5k}}$, now we can use $mg = \frac{\sqrt{3}}{2}kl$ as in previous method

thus $T = 2\pi\sqrt{\frac{\sqrt{3}l}{10g}}$ Ans.

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